

Personal Information

Address
Hanguangbeilu 10
710068, Xi'an, Shaanxi
China

Phone
+49 177-909-2448

E-Mail
shaofei.liu@shaofeiliu.com

Web
www.shaofeiliu.com

Skills

Programming
- Python (PyTorch, NumPy, Pandas)
- C++ (Algorithm Development)
- CUDA (Kernel Implementation, GPU-accelerated Computing)

Machine Learning & AI
- Deep Learning & Data-driven Modeling
- Anomaly Detection & Classification
- Explainable AI & RAG/LLM
- Model Training & Evaluation

Sensors & Data Processing
- Sensor Fusion & Multimodal Data
- Time Series & Signal Processing
- Computer Vision

Simulation & Engineering
- ML Projects on AWS
- MATLAB / Simulink
- SolidWorks, AutoCAD
- Git, LaTeX, MS Office

Languages

- Chinese (Native)
- German (C1, TestDaF 17)
- English (C1, IELTS 7.5)

Shaofei Liu

Software & AI Engineer

Research-oriented engineer focused on multimodal machine learning, sequence modeling, sensor fusion, signal processing, and explainable AI. Experienced in the modeling, development, and robust validation of intelligent systems across radar-based, medical, mobile, and safety-critical application domains. International academic and professional background in Germany and China.

Professional Experience

05/2025 – Present
Independent AI Engineer
Munich, Germany / Shanghai, China
- Modeling of visitor path analysis using GNN/RL and VR-based 3D reconstruction, with validation on real museum data.
- Knowledge-based fault diagnosis using Knowledge Graphs/RAG, and sensor fusion for decision support in flight-emergency scenarios.

03/2024 – 04/2025
Software Developer
The Pets Team GmbH & Co. KG, Grünwald, Germany
- Development and evaluation of the software architecture and interfaces of Android smart devices under OTA updates and lifecycle conditions.
- Conceptualization of automated content generation and service chatbots for DACH clients using LLM applications for process optimization.

04/2022 – 12/2023
AI Algorithm Engineer
Wisemed Medical Technology Co. Ltd, Beijing, China
- Development of CNN/Transformer models and multimodal fusion of DICOM/EHR data for diagnostic decision support.
- Establishment of image classification, explainable AI, and robustness validation to support the Medical Digital Twin in clinical data pipelines.

03/2019 – 09/2019
Algorithm Development Intern
Continental Automotive GmbH, Regensburg, Germany
- MIMO-FMCW signal processing and FFT clutter suppression with experimental validation of 2D target detection in complex scenarios.
- Conducted system integration and comparative tests to quantify detection accuracy and false-alarm rate.

Education

10/2019 – 02/2022
Master of Science in Robotics, Cognition, Intelligence
Technical University of Munich, Germany | Overall Grade: 1.8
Master's Thesis: RNNs with Independency Assumptions: Scalable and Efficient Sequence Learning | Thesis Grade: 1.0

10/2018 – 09/2019
Bachelor of Engineering in Mechatronics / Precision Engineering
Munich University of Applied Sciences, Germany
(Sino-German Double Degree Program)

09/2015 – 09/2019
Bachelor of Engineering in Mechatronics
Tongji University, China | Overall Grade: 4.62/5.0 (approx. German grade 1.7)
Bachelor's Thesis: Development of an obstacle detection algorithm using MIMO FMCW radar | Thesis Grade: 5.0/5.0 (approx. German grade 1.0)